Meeting III: Intro to Category Theory Pt. 2

Topoi: The Categorical Analysis of Logic

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May 26, 2026

Recap

Last time, we covered the definitions of a quiver, path, and category.

Intuition

A **category** is a quiver equipped with an associative, unital path reduction operation.

- The operation is expressible via a binary composition with identities.

$$
\begin{tikzcd}
a \arrow[r, "f"] & b \arrow[d, "1_b"] & & & & \\
& b \arrow[r, "g"'] & c \arrow[d, "h"] & \Longmapsto & a \arrow[dr, "h \circ g \circ f"] & \\
& & d & & & d
\end{tikzcd}
$$

Goals for today

By the end of this meeting, we should understand:

- a categorical *product*, which is:
  - a *terminal* object
  - in the category of *cones*
  - of an arrowless *diagram*
- isomorphisms
- universal properties
- and as time permits:
  - duality and the opposite category
  - monics and epics

Terminal Objects

Terminal Object: Definition

Definition

A **terminal vertex** of a quiver $Q = (V, E, s, t)$ is a vertex $T \in V$ such that:

$$(\forall v \in V)(\exists ! e \in E)(s(e) = v \wedge t(e) = T)$$

In English: every vertex (including T itself) admits a unique edge to T.

Definition

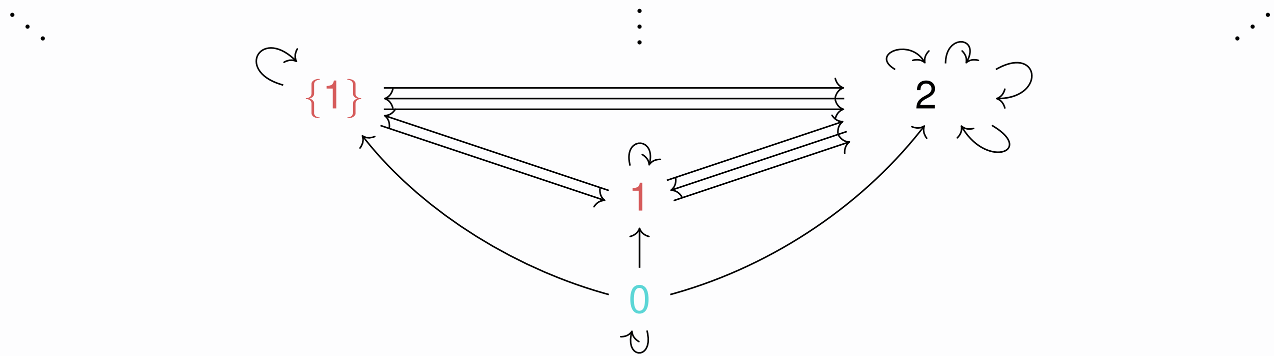
A **terminal object** of a category $\mathscr{C}$ is a terminal vertex of the underlying quiver of $\mathscr{C}$.

$$\begin{tikzcd} V \arrow[r, dashed, "!", "!_V"'] & T & \\ & & W \arrow[ul, "!_w"] \end{tikzcd}$$

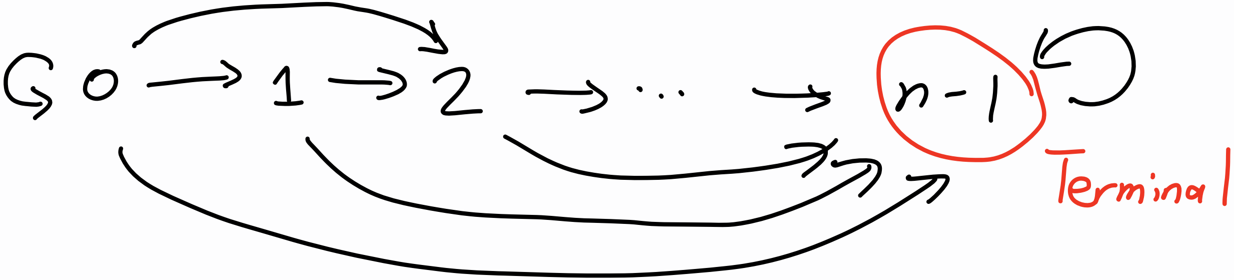
$!_V$

Example: Terminal Object in Set

$2=\{0,1\}=\{\{\},\{\{\}\}\}$



Example: Terminal Object in n for $n > 0$



Terminal Object: Non-Example

$$\begin{tikzcd} a \arrow[loop left, "1_a"] \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & b \arrow[loop right, "1_b"] \end{tikzcd}$$

Intuition

- There is no arrow from b to a.
- There is more than one arrow from a to b.

Terminal Object: Ambiguous Example

$$X$$

$$\begin{tikzcd} X \arrow[loop above, "{\langle X, X, \langle\rangle \rangle}"] \end{tikzcd}$$

Notes

- As a quiver, there is no arrow from X to itself.
- The "path category" generated by this quiver admits X as terminal.

Diagrams

Quiver Homomorphism: Definition

Let $Q_1 = (V_1, E_1, s_1, t_1)$ and $Q_2 = (V_2, E_2, s_2, t_2)$ be two quivers.

Definition

A **quiver homomorphism** f from Q_1 to Q_2 comprises:

- an assignment $f_0 : V_1 \to V_2$
- an assignment $f_1 : E_1 \to E_2$ preserving sources and targets:

$$s_2(f_1(e)) = f_0(s_1(e)) \qquad t_2(f_1(e)) = f_0(t_1(e)) \qquad \forall e \in E_1$$

$$\begin{tikzcd} E_1 \arrow[r, "f_1"] \arrow[d, "s_1"'] & E_2 \arrow[d, "s_2"] \\ V_1 \arrow[r, "f_0"'] & V_2 \end{tikzcd} \qquad \begin{tikzcd} E_1 \arrow[r, "f"] \arrow[d, "t"'] & E_2 \arrow[d, "t"] \\ V_1 \arrow[r, "f"'] & V_2 \end{tikzcd}$$

Diagram: Definition

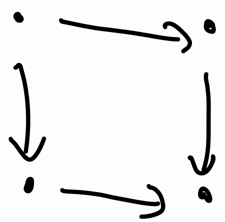
Let $\mathscr{C} = (O, A, \mathrm{dom}, \mathrm{cod}, \circ, 1)$ be a category and $Q = (V, E, s, t)$ a quiver.

Definition

A **diagram** in a category $\mathscr{C}$ of shape Q is a quiver homomorphism D from Q to the underlying quiver of $\mathscr{C}$.

$$\mathrm{dom}(D_1(e)) = D_0(s(e)) \qquad \mathrm{cod}(D_1(e)) = D_0(t(e)) \qquad \forall e \in E$$

$$\begin{tikzcd} E \arrow[r, "D"] \arrow[d, "s"'] & A \arrow[d, "\mathrm{dom}"] \\ V \arrow[r, "D"'] & O \end{tikzcd} \qquad \begin{tikzcd} E \arrow[r, "D"] \arrow[d, "t"'] & A \arrow[d, "\mathrm{cod}"] \\ V \arrow[r, "D"'] & O \end{tikzcd}$$



Example: A Span in Set

$$
\begin{tikzcd}
\text{Shape:} & 1 \arrow[d, Rightarrow] & & 0 \arrow[ll, "h_{01}"'] \arrow[rr, "h_{02}"] \arrow[d, Rightarrow] & & 2 \arrow[d, Rightarrow] \\
\text{Diagram:} & X & & \{\langle x, S\rangle : x \in S \subseteq X\} \arrow[ll, "\pi_1"] \arrow[rr, "\pi_2"'] & & \mathscr{P}(X)
\end{tikzcd}
$$

$$x \longmapsfrom \langle x, S \rangle \longmapsto S$$

Example: The Empty Diagram in $\mathscr{C}$

Shape: $(\emptyset, \emptyset, !, !)$

Diagram:

$$\emptyset \longrightarrow C_0$$

$$\emptyset \longrightarrow C_1$$

Example: An Arrowless Diagram in $\mathscr{C}$

Shape:

Diagram:

$$\begin{tikzcd} 0 \arrow[d, Rightarrow] & 1 \arrow[d, Rightarrow] \\ A & B \end{tikzcd}$$

$Q = (I, \emptyset, !, !)$

$D = (D_0, D_1)$

$\boxed{D_0 : I \longrightarrow C_0}$

$D_1 : \emptyset \longrightarrow C_1$

Commutative Diagram: Definition

Definition

A diagram $D = (D_0, D_1)$ in a category $\mathscr{C}$ is called **commutative** if parallel paths have the same composite.

$$
\begin{tikzcd}
 & A \arrow[dl, "f"'] \arrow[ddr, "\alpha"] & \\
B \arrow[d, "g"'] & & \\
C \arrow[dr, "\gamma"'] & = & X \arrow[dl, "h"] \\
 & Z &
\end{tikzcd}
$$

$\gamma \circ g \circ f = h \circ \alpha$

Example: Commutative Diagram

$$
\begin{tikzcd}
\mathbb{N} \arrow[d, equal, "1_{\mathbb{N}}"'] & \mathbb{N} \arrow[l, "k \mapsto 2k"'] \arrow[r, "!"] \arrow[d, "k \mapsto \langle 2k, 2\mathbb{N}\rangle"] & 1 \arrow[d, "2\mathbb{N}"] \\
\mathbb{N} & \{\langle x, S\rangle : x \in S \subseteq \mathbb{N}\} \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & \mathscr{P}(\mathbb{N})
\end{tikzcd}
$$

Example: Non-commutative Diagram

$$\begin{tikzcd} \mathbb{Z} \arrow[loop above, "x\mapsto x+1"] \end{tikzcd}$$

$$\begin{tikzcd} \langle \mathbb{Z}, \mathbb{Z}, \langle\rangle \rangle \arrow[d, mapsto] \\ 1_{\mathbb{Z}} \end{tikzcd}$$

Cones

Cone: Definition

Definition

A **cone** $C = (x, \langle f_i \rangle_{i \in I})$ in $\mathscr{C}$ for an arrowless diagram $D = (d, !)$ of shape $(I, \emptyset, !, !)$ is:

- a $\mathscr{C}$-object x, together with
- a $\mathscr{C}$-arrow $f_i : \cancel{x} \to d_i$ for each index $i \in I$.
  X

$$
\begin{tikzcd}
 & & & x \arrow[llld, "f_{i_0}"'] \arrow[lld, "f_{i_1}"'] \arrow[ld, phantom, "\cdots"] \arrow[rd, "f_{i_k}"] \arrow[rrd, phantom, "\cdots"] & & \\
d_{i_0} & d_{i_1} & \cdots & & d_{i_k} & \cdots
\end{tikzcd}
$$

Example: A Cone in Set

Shape:	1		2
	$\Downarrow$		$\Downarrow$
Diagram:	X		$\mathscr{P}(X)$
Cone:	X	$\xleftarrow[\pi_1]{} \{\langle x, S\rangle : x \in S \subseteq X\} \xrightarrow[\pi_2]{}$	$\mathscr{P}(X)$

Intuition

- A cone is an extended diagram over a shape that adjoins an "initial object".
  - The new loop is required to map to an identity morphism.
  - Alternatively, discard the new loop from the shape.

Cone Morphism: Definition

Let $(x, \langle f_i \rangle_{i \in I})$ and $(y, \langle g_i \rangle_{i \in I})$ be cones for an arrowless diagram D with vertices I.

Definition

A **cone morphism** from $(x, \langle f_i \rangle_{i \in I})$ to $(y, \langle g_i \rangle_{i \in I})$ is a $\mathscr{C}$-arrow $h : x \to y$ such that the following diagram is commutative:

$$\begin{tikzcd}
 & & & x \arrow[dlll, "f_{i_0}"'] \arrow[dll, "f_{i_1}"'] \arrow[dr, "f_{i_k}"] \arrow[dd, "h"] & & \\
D(i_0) & D(i_1) & \cdots & & D(i_k) & \cdots \\
 & & & y \arrow[ulll, "g_{i_0}"] \arrow[ull, "g_{i_1}"] \arrow[ur, "g_{i_k}"'] & &
\end{tikzcd}$$

$\forall i_k \in I$

$g_{i_k} \circ h = f_{i_k}$

Example: Cone Morphism

$$e^{ix} = \cos(x) + i\sin(x)$$

$$\begin{tikzcd}
 & \mathbb{R} \arrow[dl, "\cos(x)"'] \arrow[dr, "\sin(x)"] \arrow[dd, "e^{ix}"] & \\
\mathbb{R} & & \mathbb{R} \\
 & \mathbb{C} \arrow[ul, "\operatorname{Re} z"] \arrow[ur, "\operatorname{Im} z"'] &
\end{tikzcd}$$

Example: Identity Cone Morphism

$(x, \langle f_i \rangle_i)$

$$
\begin{tikzcd}
 & & & x \arrow[dlll, "f_{i_0}"'] \arrow[dll, "f_{i_1}"'] \arrow[dd, "1_x"] \arrow[dr, "f_{i_k}"] & & \\
d_{i_0} & d_{i_1} & \cdots & & d_{i_k} & \cdots \\
 & & & x \arrow[ulll, "f_{i_0}"] \arrow[ull, "f_{i_1}"] \arrow[ur, "f_{i_k}"'] & &
\end{tikzcd}
$$

Example: Composite Cone Morphism

$$\begin{tikzcd}
(x, \langle f_i \rangle_i) \arrow[r, "\alpha"] \arrow[dr, "\beta \circ \alpha"'] & (y, \langle g_i \rangle_i) \arrow[d, "\beta"] \\
& (z, \langle h_i \rangle_i)
\end{tikzcd}$$

$$\begin{tikzcd}
 & & & x \arrow[dll, "f_{i_1}"'] \arrow[dr, "f_{i_k}"] \arrow[dd, "\alpha"] & & \\
\cdots & d_{i_1} & \cdots & & d_{i_k} & \cdots \\
 & & & y \arrow[ull, "g_{i_1}"] \arrow[ur, "g_{i_k}"'] \arrow[d, "\beta"] & & \\
 & & & z \arrow[uull, "h_{i_1}"] \arrow[uur, "h_{i_k}"'] & &
\end{tikzcd}$$

Product: Definition

Cones for a given diagram form a category whose arrows are the cone morphisms.

Definition

A **product** is a terminal object in the category of cones for an arrowless diagram.

“the”

$$
\begin{tikzcd}
 & & & X \arrow[dlll, "f_{i_0}"'] \arrow[dll, "f_{i_1}"'] \arrow[dl, "\cdots"'] \arrow[dr, "f_{i_k}"] \arrow[drr, "\cdots"] \arrow[dd, dashed, "\exists!\langle f_i \rangle_i"] & & \\
d_{i_0} & d_{i_1} & \cdots & & d_{i_k} & \cdots \\
 & & & \prod\limits_{i \in I} d_i \arrow[ulll, "\pi_{i_0}"] \arrow[ull, "\pi_{i_1}"] \arrow[ul, "\cdots"] \arrow[ur, "\pi_{i_k}"'] \arrow[urr, "\cdots"'] & &
\end{tikzcd}
$$

Example: Cartesian Product in Set

$$\begin{tikzcd} & x \arrow[dl, "f"'] \arrow[dr, "g"] \arrow[dd, dashed, "\langle f, g \rangle"] & \\ a & & b \\ & a \times b \arrow[ul, "\pi_1"] \arrow[ur, "\pi_2"'] & \end{tikzcd}$$

If $e \in x$,

then

$\langle f, g \rangle (e)$

$= \langle f(e), g(e) \rangle$

Example: Products in Posets $(P, \leq)$

$a, b \in P$

A product of a & b, if it exists, is an $c \in P$ s.t.

$$
\begin{tikzcd}
 & x \arrow[dl, "\leq"'] \arrow[dr, "\leq"] \arrow[dd, dotted, "\leq"] & \\
a & & b \\
 & c \arrow[ul, "\leq"', "\pi_1"] \arrow[ur, "\leq", "\pi_2"'] &
\end{tikzcd}
$$

c is a g.l.b.

inf

Non-Example: Ordinal Product

Ord

$$2 \cdot \omega = \omega$$

$$\omega \cdot 2 = \omega + \omega \neq \omega$$

Tensor Product

Smash Product

Isomorphisms

Isomorphism: Definition

Definition

An **isomorphism** (abbreviated as **iso**) is an arrow $f : a \to b$ such that:

$$(\exists g : b \to a)(g \circ f = 1_a \wedge f \circ g = 1_b)$$

In English: f has a two-sided inverse.

Intuition

The inverse g is the unique morphism making the following diagram commutative:

$$\begin{tikzcd} a \arrow[r, shift left, "f"] & b \arrow[l, shift left, dashed, "g"] \end{tikzcd}$$

$$\begin{tikzcd} & b \arrow[dr] & \\ a \arrow[ur] & & c \arrow[ll] \end{tikzcd}$$

Example: Isomorphisms

$$\begin{tikzcd}
\{-1,1\} \arrow[r, shift left, "(1-x)/2"] \arrow[r, bend left=50, "(1+x)/2"] & \{0,1\} \arrow[l, shift left, "(-1)^x"] \arrow[l, bend left=50, "2x-1"]
\end{tikzcd}$$

Universal Property: Definition

Definition

- A **universal property** is a characterization of a construction up-to-unique morphism.
- This is a fancy way of saying that a construction is a terminal object in a category.

Notes

- That something is not necessarily unique.

Theorem: Universal Constructions Are Isomorphic

Theorem

Any two are isomorphic to each other via the unique morphism

T_1, T_2 are terminal $\mathcal{C}$-objects

$$\Rightarrow \begin{tikzcd} T_1 \arrow[loop left, "1_{T_1}"] \arrow[r, bend right, "g"'] & T_2 \arrow[l, bend right, "f"'] \arrow[loop right, "1_{T_2}"] \end{tikzcd}$$

Does $f \circ g = 1_{T_1}$? ✓

$g \circ f = 1_{T_2}$ ✓

Example: Universal Property of Products

$A \qquad B$

$$\begin{tikzcd}
 & X \arrow[dl, bend right, "f"'] \arrow[dr, bend left, "g"] \arrow[d, dashed, "\langle f, g\rangle"] & \\
A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] \arrow[d, bend left, "\langle \pi_1, \pi_2\rangle_P"] & B \\
 & P \arrow[ul, bend left, "p_1"] \arrow[ur, bend right, "p_2"'] \arrow[u, bend left, "\langle p_1, p_2\rangle_{A\times B}"] &
\end{tikzcd}$$

$$\langle p_1, p_2\rangle_{A\times B} = \langle \pi_1, \pi_2\rangle_P^{-1}$$

$|A| \geq 2$ $\qquad$ $swap \neq 1_{A\times A}$

$$\begin{tikzcd}
A & A \times A \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] \arrow[d, dashed, bend right, "swap'"'] & A \\
 & A \times A \arrow[ul, bend left, "\pi_2"] \arrow[ur, bend right, "\pi_1"'] \arrow[u, dashed, bend right, "swap"'] &
\end{tikzcd}$$

Duality

Opposite Category: Definition

Let $\mathscr{C} = (O, A, \mathrm{dom}, \mathrm{cod}, \circ, 1)$ be a category.

Definition

The **opposite category** of $\mathscr{C}$ is $\mathscr{C}^{\mathrm{op}} = (O, A, \mathrm{cod}, \mathrm{dom}, \circ', 1)$, where

$$f \circ' g = g \circ f$$

for all $\mathscr{C}$-composable pairs $a \xrightarrow{f} b \xrightarrow{g} c$.

Notation

- If $f : a \to b$, then $f^{\mathrm{op}} : b \to a$
- If x is a $\mathscr{C}$-object, then x is also a $\mathscr{C}^{\mathrm{op}}$-object.

Duality Principle

Definition

Given a sentence Σ expressible in the (two-sorted) language of categories, the *dual* sentence Σ^{op} is obtained by:

- interchanging dom and cod
- replacing $\circ : A\,{}_{\mathsf{dom}}\!\times_{\mathsf{cod}}\,A \to A$ with $\circ' : A\,{}_{\mathsf{cod}}\!\times_{\mathsf{dom}}\,A \xrightarrow{\mathsf{swap}} A\,{}_{\mathsf{dom}}\!\times_{\mathsf{cod}}\,A \xrightarrow{\circ} A$

Theorem

If a sentence Σ is provable from the axioms of a category, then so is the sentence Σ^{op}.

Example: Dual of Terminal Object

$$\varphi(T) := (\forall x \in \mathcal{C}_0)(\exists! f \in \mathcal{C}_1)(\mathrm{dom}(f) = x \wedge \mathrm{cod}(f) = T)$$

$$\varphi^{op}(I) := (\forall x \in \mathcal{C}_0)(\exists! f \in \mathcal{C}_1)(\mathrm{cod}(f) = x \wedge \mathrm{dom}(f) = I)$$

<u>Initial Object</u>

In Set: $\emptyset$

Example: Dual of a Monic

$$\forall c \underset{\forall h}{\overset{\forall g}{\rightrightarrows}} a$$

Definition

An arrow $f : a \to b$ is called **monic** if it is left-cancellable: $f \circ g = f \circ h$ implies $g = h$

In **Set**, a function is monic iff it is injective.

Definition$^{\mathrm{op}}$

An arrow $f : a \to b$ is called **epic** if it is right-cancellable: $g \circ f = h \circ f$ implies $g = h$

In **Set**, a function is epic iff it is surjective.

$$b \underset{\forall h}{\overset{\forall g}{\rightrightarrows}} \forall d$$

Exercise: Cones for The Empty Diagram in $\mathcal{C}$

Shape:

Diagram:

$$\begin{tikzcd}
(X, \langle\rangle) \arrow[d, "f"'] \arrow[dr, dashed] & \\
& (T, \langle\rangle) \\
(Y, \langle\rangle) \arrow[ur, dashed] &
\end{tikzcd}$$

cone $\leftrightarrow$ cocone

product $\leftrightarrow$ coproduct

terminal $\leftrightarrow$ initial

"the"

Exercise Review

Outro

Reading for Next Lecture

- Minimal
  - Sections 10 and 13 of chapter 3
  - Sections 1 and 2 of chapter 4
- Excellent
  - All of chapter 3
  - Sections 1 through 4 of chapter 4
- Extra
  - Limits and Colimits